

Set up for Calculations:

A matrix is a mathematical object consisting of numbers in cells, we will look at 2×2 and 3×3 matrices

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

2×2

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

3×3

An important calculation with matrices is a determinant

For a 2×2 matrix the determinant is found via

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

For a 3×3 we use something called Co-factor expansion

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$= a(ei - hf) - b(di - gf) + c(dh - ge)$$

This is useful when calculating something called the cross product.

$$\text{Let } \vec{a} = \langle a_1, a_2, a_3 \rangle, \quad \vec{b} = \langle b_1, b_2, b_3 \rangle$$

Let's use the notation

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}, \quad \vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$$

Then, the cross product is

$$\begin{aligned} \vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \vec{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \vec{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \vec{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \\ &= \vec{i}(a_2b_3 - b_2a_3) - \vec{j}(a_1b_3 - b_1a_3) + \vec{k}(a_1b_2 - b_1a_2) \end{aligned}$$

$$= \langle a_2b_3 - b_2a_3, b_1a_3 - a_1b_3, a_1b_2 - b_1a_2 \rangle$$

Notice the result is a vector.

Ex: $\vec{a} = \langle 1, 3, 4 \rangle, \quad \vec{b} = \langle 2, 7, -5 \rangle$

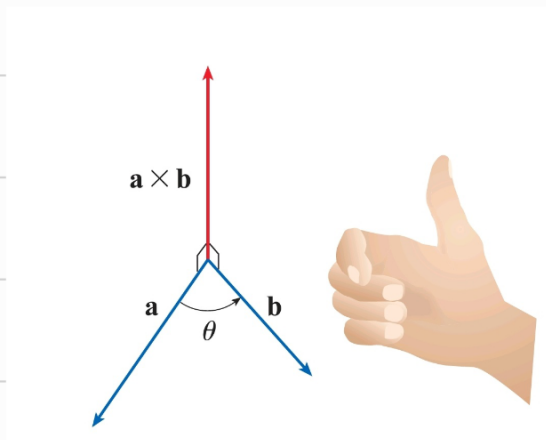
$$\begin{aligned} \vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 3 & 4 \\ 2 & 7 & -5 \end{vmatrix} = \vec{i} \begin{vmatrix} 3 & 4 \\ 7 & -5 \end{vmatrix} - \vec{j} \begin{vmatrix} 1 & 4 \\ 2 & -5 \end{vmatrix} + \vec{k} \begin{vmatrix} 1 & 3 \\ 2 & 7 \end{vmatrix} \\ &= \vec{i}(3(-5) - 7 \cdot 4) - \vec{j}(1(-5) - 2 \cdot 4) + \vec{k}(1 \cdot 7 - 2 \cdot 3) \\ &= \langle -43, 13, 1 \rangle \end{aligned}$$

The important result of this is that

Thm:

The vector $\vec{a} \times \vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$.

For direction, use right hand rule.



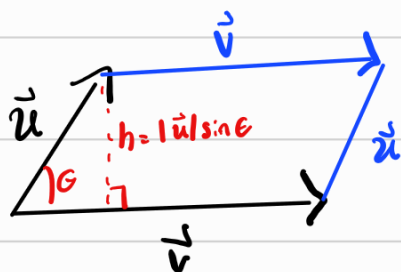
Thm:

If θ is the angle between $\vec{a}$ and $\vec{b}$,
then

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin(\theta)$$

$\rightarrow \vec{a}$ and $\vec{b}$ are parallel if and only if $\vec{a} \times \vec{b} = \vec{0}$

Geometry



$$\begin{aligned} \text{Area} &= b \cdot h = |\vec{v}| \cdot h = |\vec{v}| |\vec{u}| \sin \theta \\ &= |\vec{u} \times \vec{v}| \end{aligned}$$

Ex: Find a vector perpendicular to the plane that passes through $P(1, 4, 6)$ $Q(-2, 5, -1)$ $R(1, -1, 1)$

$$\vec{PQ} = \langle -3, 1, -7 \rangle \quad \vec{PR} = \langle 0, -5, -5 \rangle$$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} i & j & k \\ -3 & 1 & -7 \\ 0 & -5 & -5 \end{vmatrix}$$

$$= i \begin{vmatrix} 1 & -7 \\ -5 & -5 \end{vmatrix} - j \begin{vmatrix} -3 & -7 \\ 0 & -5 \end{vmatrix} + k \begin{vmatrix} -3 & 1 \\ 0 & -5 \end{vmatrix}$$

$$= i(-5 - 35) - j(15) + k(15)$$

$$\boxed{= \langle -40, -15, 15 \rangle}$$

P, Q, R form a triangle w/ area

$$\frac{1}{2} |\vec{PQ} \times \vec{PR}| = \frac{1}{2} \sqrt{(40)^2 + (-15)^2 + (15)^2} = \frac{5}{2} \sqrt{82}$$

Properties of Cross Product

Let $\vec{a}, \vec{b}, \vec{c}$ be vectors, $c \in \mathbb{R}$.

- 1) $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$
- 2) $c \vec{a} \times \vec{b} = c(\vec{a} \times \vec{b}) = \vec{a} \times c\vec{b}$
- 3) $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$
- 4) $(\vec{a} + \vec{b}) \times \vec{c} = \vec{a} \times \vec{c} + \vec{b} \times \vec{c}$
- 5) $\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c}$
- 6) $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$